

Project Report: Climate and Respiratory Health Analysis

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1 Introduction

Climate change has been widely discussed throughout the world over the last few decades, and it is now recognised as a fundamental threat to human health. It affects the physical environment as well as all aspects of both natural and human systems, but just how much does it really affect an everyday citizen? In this project, we aim to address this exact question, namely through analysing the links between meteorology, air pollution, and respiratory health worldwide on a monthly and yearly time scale.

2 Problem Definition and Scope

2.1 Problem Definition

The main goal of this project is to analyse how different meteorological and air pollution variables affect respiratory health in various countries worldwide. It is important to note that respiratory conditions such as COPD, asthma, and lung cancer manifest differently across individuals. The same disease can range from mild symptoms to fatal outcomes due to differences in lifestyle, genetics, or environmental exposure. In this work, we take a simple metric for health outcomes – **mortality rates** for each disease. Therefore, the core objective of the research can be formulated as:

- Can we establish quantitative associations between air pollutants, climate conditions, and respiratory disease mortality across countries?

To be more specific, we will also analyse the following:

- Which environmental factors ($PM_{2.5}$, PM_{10} , NO_2 , O_3 , SO_2 , CO , temperature, humidity) are most strongly linked to respiratory mortality?
- Which variables exactly are highly correlated with each disease?
- Can we see differences in mortality rates between the female and male populations? Can this be explained by our analysis?
- How are countries clustered based on their exposure to air pollutants?
- Which climate variables are the main drivers in our analysis?

2.2 Scope and Data Acquisition Strategy

This study focusses on analysing the relationship between climate and respiratory health at the population level on a *global scale*. The analysis covers nearly 180 countries spanning Europe, Asia, Africa, and the Americas, allowing for broad geographic comparison and variability in both environmental conditions and health outcomes. The **temporal scope** of the study spans the years 1980-2023, with data aggregated at monthly, seasonal, and yearly resolutions to capture both short-term variability and long-term trends.

We differentiate between the **input** (environmental and meteorological variables) and **output** data (health outcomes), as they were gathered in a different manner from different sources. In this section, we will detail the steps for acquiring our data.

2.2.1 Environmental and Meteorological Data (Inputs)

For climate data, we used the following two sources:

1. Copernicus Climate Data Store – ERA5 monthly averaged data

Meteorological features capture weather patterns and extreme conditions that influence pollutant behaviour and health stressors. From this dataset, we use meteorological variables such as temperature, relative humidity, wind speed, and surface pressure.

Key Variables Acquired:

- **t2m**: 2-metre air temperature (K).
- **d2m**: 2-metre dewpoint temperature (K) – utilised to derive relative humidity.
- **tp**: Total precipitation (m).
- **sp**: Surface pressure (Pa).
- **u10, v10**: 10-metre wind components (m/s).

Temporal Scope: 1980-2023, monthly averages.

2. Copernicus Atmosphere Monitoring Service (CAMS)

This dataset provides information about key air pollutants: PM_{2.5}, NO₂, O₃. We choose this dataset as these pollutants serve as the main explanatory variables representing short- and long-term exposure to air pollution.

Variables Acquired:

- **pm2p5**: Fine particulate matter (PM_{2.5}) – particles $\leq 2.5\mu\text{m}$ in diameter.
- **pm10**: Coarse particulate matter (PM₁₀) – particles $\leq 10\mu\text{m}$ in diameter.
- **tcno2**: Total column nitrogen dioxide (NO₂).
- **gtco3**: Total column ozone (O₃).
- **tcs02**: Total column sulfur dioxide (SO₂).
- **tcco**: Total column carbon monoxide (CO).

Temporal Scope: As this is a relatively new dataset, we only managed to gather data covering the years 2003-2023.

Collection Method: Data was downloaded via the Copernicus Atmosphere Data Store (ADS)¹ and Climate Data Store (CDS)² APIs in NetCDF format. The raw files contain global gridded data at approximately 0.75° spatial resolution with monthly temporal resolution. Details about how we handled the gridded data and transformed it into human readable tabular data will be discussed in the next section.

2.2.2 Health Outcome Data (Outputs)

Health data was sourced from the Institute for Health Metrics and Evaluation (IHME)³.

1. Global Burden of Disease (GBD)

The GBD study provides standardised estimates of mortality rates (per 100,000 population) for specific causes, stratified by country, year, and sex.

Disease Variables Acquired:

- Chronic Obstructive Pulmonary Disease (COPD)
- Asthma
- Lower Respiratory Infections (LRI)

Temporal Scope: 1980-2023, although important to note that not all countries have data spanning all these years.

¹<https://ads.atmosphere.copernicus.eu/>

²<https://cds.climate.copernicus.eu/>

³<https://www.healthdata.org/>

3 Data Processing

The raw data acquired from Copernicus (ERA5 and CAMS) and the IHME (GBD) possess fundamentally different structures. The meteorological and pollution data are **spatially continuous** (gridded) and temporally high-resolution (monthly), whereas the health data are **spatially discrete** (country-level) and temporally low-resolution (yearly). This section details the methodology used to harmonise these **input datasets** into a unified structure suitable for statistical analysis.

3.1 Spatial Aggregation: From Grid to Regional Level

The primary challenge in integrating the environmental inputs was converting gridded satellite reanalysis data into country-level metrics.

The ERA5 and CAMS datasets are provided in NetCDF format with a spatial resolution of approximately $0.25^\circ \times 0.25^\circ$ (latitude $\times$ longitude). To align this with the health data, we performed the following steps:

1. **Masking:** We loaded the NetCDF files using the `xarray` library. We then utilised the `regionmask` library in conjunction with Natural Earth shapefiles to create boolean masks for the administrative boundaries of all 180 countries.
2. **Area-Weighted Averaging:** Simple arithmetic means can be biased due to the converging of longitudinal lines at the poles (grid cells are smaller at higher latitudes). To correct for this, we computed the **area-weighted spatial average** for all grid points falling within a country's boundary.
3. **Extraction:** This process reduced the 3D global grids into 2D time-series vectors (Country $\times$ Time) for every pollutant and meteorological variable.

3.2 Temporal Alignment and Resampling

The health outcome data (GBD) is reported annually. To enable correlation and regression analysis, the meteorological inputs required temporal downsampling.

- **Annual Aggregation:** We aggregated the monthly environmental data into yearly averages to match the GBD resolution. For temperature and pollutants, we calculated the arithmetic mean. To retain important information, i.e. extreme weather conditions/peaks in summer or winter, which we hypothesise have direct effect on respiratory health, we also computed different metrics for some variables: **std**, **min/max** (capturing extremes), **variance**, etc.
- **Seasonality Preservation:** Although direct correlation with annual mortality requires yearly aggregation, we decided to also keep seasonal trends. E.g., higher NO_2 in winter or Ozone spikes in summer that might drive acute respiratory exacerbations.

3.3 GBD Data Structuring

The raw GBD data was provided in a "long" format. We pivoted this to a "wide" format to create distinct features for each disease and sex.

- **Pivoting:** Each unique combination of **cause** and **sex** was transformed into its own column.
- **Naming Convention:** Features were renamed following the pattern `disease_statistic_sex` (e.g., `asthma_mortality_F`).
- **Imputation:** Missing values in the health records were imputed using GBD modeled estimates, cross-validated against WHO mortality database records where available to ensure consistency.

4 Feature Engineering

To enhance the predictive capacity of our models and extract latent patterns from the high-dimensional environmental data, we implemented a comprehensive feature engineering pipeline. This process involved domain-specific variable derivation, seasonal signal extraction, and unsupervised dimensionality reduction techniques. All transformations were performed using Python (Pandas, Scikit-Learn) prior to the final statistical analysis.

4.1 Meteorological Derivations and Unit Conversion

The raw ERA5 reanalysis data is provided in SI base units or vector components, which are not directly intuitive for health impact assessments. We transformed these variables into metrics with established physiological relevance:

- **Temperature Standardization:** All temperature variables ($t2m$, $d2m$, skt) were converted from Kelvin to degrees Celsius ($^{\circ}C$). This aligns the data with standard medical literature regarding heat stress and cold spells.
- **Dewpoint Depression (ΔT_{dew}):** We calculated the difference between the 2-meter air temperature and the dewpoint temperature.

$$\Delta T_{dew} = T_{air} - T_{dewpoint} \quad (1)$$

This variable serves as a physical proxy for *dryness*. Large depressions indicate low relative humidity, a known environmental trigger for bronchoconstriction in asthmatic patients and mucous membrane irritation.

- **Wind Speed Magnitude:** The ERA5 dataset provides wind as orthogonal vector components (u_{10} , v_{10}). We computed the scalar wind magnitude to represent general air movement:

$$\text{Wind}_{mag} = \sqrt{u_{10}^2 + v_{10}^2} \quad (2)$$

Wind magnitude is a critical factor in air quality modeling as it dictates the dispersion potential of pollutants. Low wind speeds (stagnation) often correlate with the accumulation of $\text{PM}_{2.5}$ and NO_2 .

4.2 Seasonal Signal Extraction

A significant limitation of using annual means for health analysis is the smoothing of extreme events. Respiratory mortality is often driven by acute seasonal stressors—such as winter influenza outbreaks exacerbated by cold, or summer cardiovascular strain caused by heatwaves. To capture these signals, we engineered season-specific features:

1. **Temporal Segmentation:** We mapped months to standard meteorological seasons (Winter: Dec-Feb; Spring: Mar-May; Summer: Jun-Aug; Autumn: Sep-Nov).
2. **Extreme Value Aggregation:** Instead of simple averages, we computed targeted metrics for relevant seasons:
 - **Winter Severity Index:** Defined as the average minimum temperature during winter months. This feature targets cold-induced respiratory stress.
 - **Summer Heat Stress Index:** Defined as the average maximum temperature during summer months, targeting heat-aggravated respiratory failure.

4.3 Dimensionality Reduction: Dual-Track PCA

A critical challenge in our dataset was the discrepancy in temporal coverage: while meteorological data (ERA5) spans the full period (1980-2023), pollution data (CAMS) is only available from 2003 onwards.

Running a single PCA on the combined dataset would make it necessary to drop data before 2003, and this would discard 23 years of valuable climate history. To address this, we implemented a **stratified PCA approach**, conducting two independent dimensionality reduction tracks.

Preprocessing: For both tracks, variables were standardized using Z-score scaling (StandardScaler) prior to analysis. This ensures that variables with large magnitudes (e.g., Pressure in Pa) do not dominate the variance calculation over trace gases.

Track 1: Meteorology (1980–2023). We analyzed 11 meteorological variables to capture global weather patterns. The first three principal components explain approximately **76%** of the total variance. Based on the eigenvector loadings (Table 1), we interpret these components as follows:

- **PC_Met_1 (Radiative-Thermal Mode):** Characterized by high solar radiation, UV, and temperature, inversely related to humidity. Positive scores indicate clear, hot, dry conditions.
- **PC_Met_2 (Hydrological Mode):** dominated by precipitation totals and relative humidity. Positive scores indicate wet, saturated conditions.
- **PC_Met_3 (Aerodynamic Mode):** Driven almost exclusively by wind speed and wind gusts, representing atmospheric turbulence independent of temperature.

Track 2: Air Pollution (2003–2023). We analyzed 6 key pollutants (PM_{2.5}, PM₁₀, NO₂, O₃, SO₂, CO). The top three components explain **84.2%** of the variance, revealing distinct chemical regimes:

- **PC_Pol_1 (Particulate Dominant):** Explaining 45.0% of variance, this component loads heavily on PM_{2.5} and PM₁₀. It serves as a proxy for dust events and solid fuel combustion.
- **PC_Pol_2 (Photochemical Oxidants):** Explaining 25.1% of variance, this is driven by NO₂ and O₃. This likely captures traffic-related emissions and secondary smog formation.
- **PC_Pol_3 (Trace Gas Combustion):** Explaining 14.1% of variance, this component is dominated by CO, likely distinguishing incomplete combustion sources from sulfur-heavy industrial emissions.

Table 1: Dominant drivers for the primary Principal Components.

Component	Primary Positive Drivers	Primary Negative Drivers
PC_Met_1	Solar Rad, UV, Temp	Relative Humidity
PC_Met_2	Precip, Humidity	Dewpoint Depression
PC_Met_3	Wind Speed, Gust	(None strong)
PC_Pol_1	PM _{2.5} , PM ₁₀	(None strong)
PC_Pol_2	O ₃ , NO ₂	PM ₁₀ , PM _{2.5}

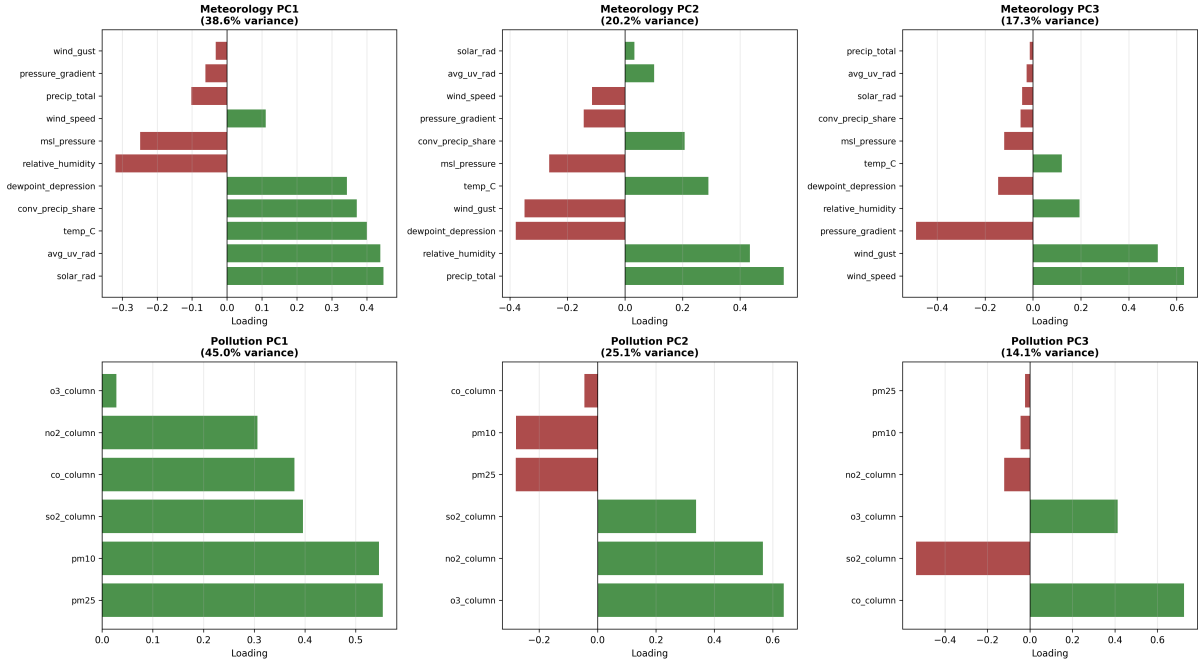


Figure 1: Explained variance ratio and cumulative variance for Meteorology and Pollution tracks.

This dual-track method allows us to utilize the full 43-year history for climate-driven mortality analysis while retaining a high-fidelity pollution model for the modern era.

4.4 Unsupervised Clustering: Climate Regimes

Grouping countries solely by geography (e.g., "Southern Europe") is often insufficient for environmental analysis, as climate zones do not strictly adhere to political borders. To address this, we employed Unsupervised Learning to discover data-driven climate regimes.

Algorithm : we used the **K-Means Clustering** algorithm on the top 5 Meteorological Principal Components (explaining $\sim 83\%$ of variance). To capture long-term climatic identity rather than short-term weather noise, we aggregated data into **Country-Decade Profiles** (e.g., "France-1980s", "France-1990s").

The objective was to partition the n profiles into $k = 6^4$ distinct clusters (S) minimizing the within-cluster sum of squares.

Stability & Climate Shifts : the clustering revealed high temporal stability: **96.7%** of country-decade profiles remained in their assigned cluster from the 1980s to the 2020s. This confirms that the clusters represent robust, **distinct climate zones**.

However, **the 3.3%** of cases that did switch clusters serve as strong indicators of shifting climate boundaries. Notably:

- **Mediterranean Warming: Spain and Greece** shifted from Cluster 4 (Temperate) to Cluster 0 (likely Semi-Arid/Warmer) in the 2020s, indicating that heat/dryness became more intense in Southern Europe.
- **Tropical Transitions: Cameroon and Côte d'Ivoire** shifted from Cluster 2 to Cluster 0, suggesting a significant change in the tropical hydrological cycle.

This data-driven stratification allows us to control for environmental context. In the subsequent mortality analysis, we can compare health outcomes across countries with **physically identical climates**, ensuring that observed differences are attributable to pollution or policy rather than weather.

⁴ $k = 6$ value was chosen as optimal after trying different values 3..7

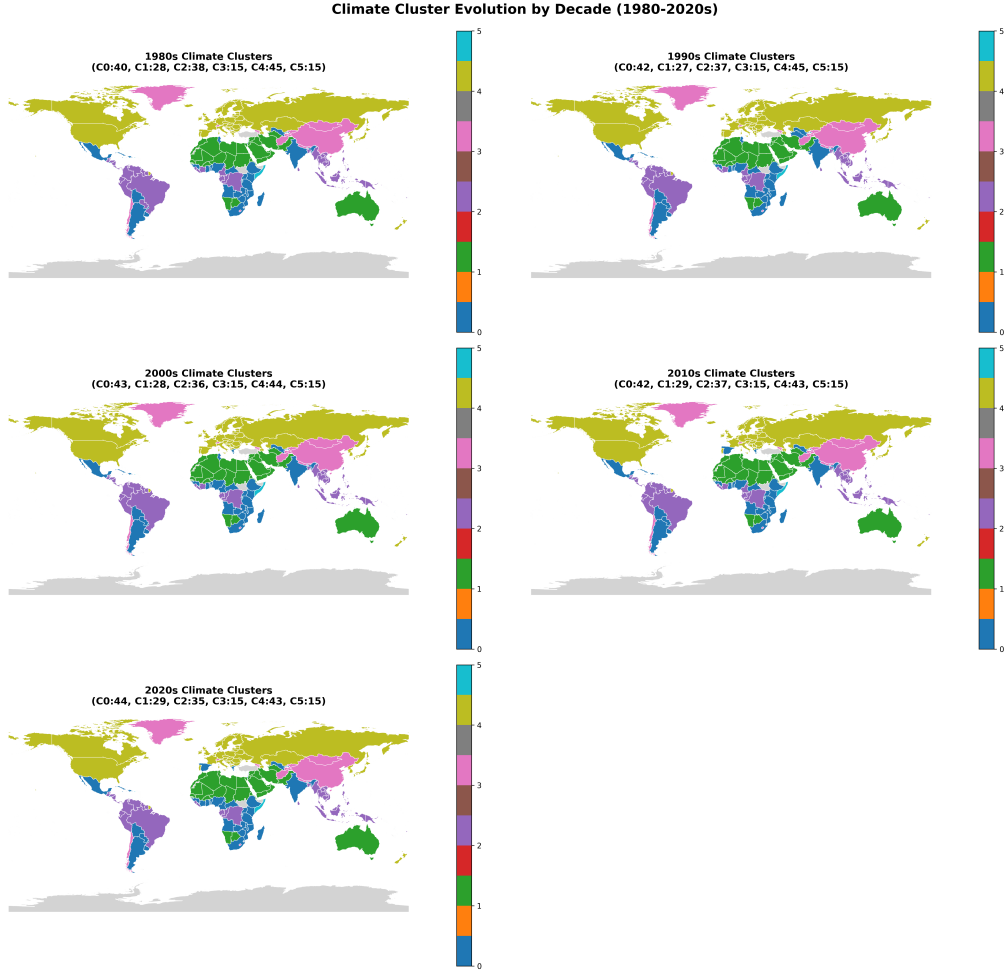


Figure 2: Clusters of countries at each decade.

4.5 Final Dataset Compilation

Following feature extraction, the dataset was aggregated to the **Country-Year** level to align with the temporal resolution of the mortality data. Redundant index permutations were removed (e.g., ensuring winter severity variables were not calculated for non-winter timestamps).

The final master dataset consists of a matrix of shape (7964, 60), representing unique Country-Year observations. The 60 feature columns are categorized as follows:

- **Raw Meteorological Aggregates:** Yearly mean, max, min, and standard deviation for temperature, precipitation, humidity, and wind vectors (15 variables).
- **Pollution Statistics:** Aggregated means and peak values for PM_{2.5}, PM₁₀, O₃, NO₂, SO₂, and CO (11 variables).
- **Principal Component Scores:** To capture complex environmental regimes, we included the yearly means and extreme values (max/min) for the 5 Meteorological PCs and 3 Pollution PCs derived in the dimensionality reduction step (14 variables).
- **Climate Clusters:** The discrete ClusterID assigned via K-Means, serving as a categorical regressor for climate type (1 variable).
- **Seasonal Indices:** Specific seasonal aggregations, including `winter_severity` and `summer_heat_stress` computed specifically for their respective seasons (16 variables).

This compiled structure ensures that the downstream regression models have access to both granular raw metrics and high-level latent environmental features.

5 Modeling Climate and Pollution Drivers of Respiratory Mortality

5.1 Methodology

To get the effects of meteorological and pollutant variables on respiratory mortality rates, we implemented a multi-stage modeling strategy. Our approach was designed to address three specific challenges that is usually seen in aggregated country-level data: **multicollinearity**, **temporal data availability** and **spatial data leakage**.

5.1.1 Study Design and Data Partitioning

Due to the discrepancy in our data between historical weather records (ERA5, available from 1980) and satellite pollution records (CAMS, available consistently from ~2003), we decided to conduct these two parallel sub-studies:

- **Study 1: The Long-Term Climate Era (1980–2023)**
 - **Scope:** Analyzed the effect of meteorological variables (Temperature, Precipitation, Humidity) over a 43-year period.
 - **Constraint:** Pollution variables were strictly excluded to prevent data loss.
- **Study 2: The Modern Pollution Era (2003–2023)**
 - **Scope:** Analysed the interaction between meteorological variables and atmospheric pollutants (PM2.5, Ozone, NO2, CO).
 - **Hypothesis:** We hypothesised that adding pollution data would significantly increase predictive accuracy for acute respiratory conditions.

5.1.2 Feature Engineering Tracks

To handle high dimensionality and multicollinearity, we defined two distinct feature sets (“Tracks”) for each study:

- **Track A: Interpretability (Raw Features)**
 - Used seasonal aggregations (e.g., `temp_C_mean_winter`, `precip_total_sum_summer`) and raw pollutant concentrations.
 - **Goal:** To identify specific, physically explainable drivers (e.g., “Winter cold drives pneumonia”).
- **Track B: Dimensionality Reduction (PCA)**
 - Used Principal Component Analysis scores derived from meteorological and pollutant blocks (`PC_Met_1`, `PC_Po1_1`).
 - **Goal:** To maximise predictive power by capturing complex, non-linear variance structures, at the cost of direct interpretability.

5.1.3 Model Selection and Validation Strategy

We decided to use a **Random Forest Regressor** (100 estimators) as our primary model due to its ability to capture non-linear interactions (for example, the compounding effect of heat and humidity). To ensure the results were robust and not artefacts of overfitting, we iterated through three validation approaches:

1. Approach 1: Random Split (Baseline)

- **Method:** Standard 80/20 train-test split (`random_state=42`).
- **Risk:** High risk of data leakage. The model likely “memorised” country-specific mortality rates (e.g., learning that “France” has low mortality) rather than learning from climate variables.

2. Approach 2: Group K-Fold (Strict Spatial Holdout)

- **Method:** Data was split by `country_name`. If a country was in the Test set, *no data* from that country (from any year) appeared in the Training set.
- **Risk:** Potential for “Covariate Shift.” If the model trained only on Temperate countries and tested on a Tropical country, it might fail as it has not been exposed to that climate zone.

3. Approach 3: Stratified Group K-Fold with Temporal Control (Final Model)

- **Method:** We used **Stratified Group K-Fold** (stratified by `cluster`), ensuring that both Training and Test sets contained a representative mix of climate zones (Tropical, Arid, Temperate) while strictly maintaining country independence.
- **Robustness Check:** We added `Year` as a control variable to account for secular trends (e.g., improvements in healthcare). This ensures that any remaining signal is attributable to environmental factors, not just the passage of time.

5.2 Results

5.2.1 Validation of the Modeling Strategy

The comparison between our three validation approaches provides a clear demonstration of the “Spatial Leakage” phenomenon often overlooked in aggregated epidemiological studies.

As illustrated in Figure 3, Approach 1 (Random Split) yielded R^2 scores exceeding 0.90 across all targets. However, the rigorous Approach 3 (Stratified Group K-Fold) revealed the true explanatory power of environmental variables to be between 0.30 and 0.44. The drop in performance confirms that the baseline model was utilizing country-specific identifiers (implicit in the data structure) rather than learning universal environmental physics.

Crucially, the fact that Approach 3 maintains an $R^2 \approx 0.44$ for LRI implies that nearly **half of the variance in lower respiratory death rates between countries** can be explained solely by climate and pollution data, without knowledge of healthcare infrastructure or GDP.

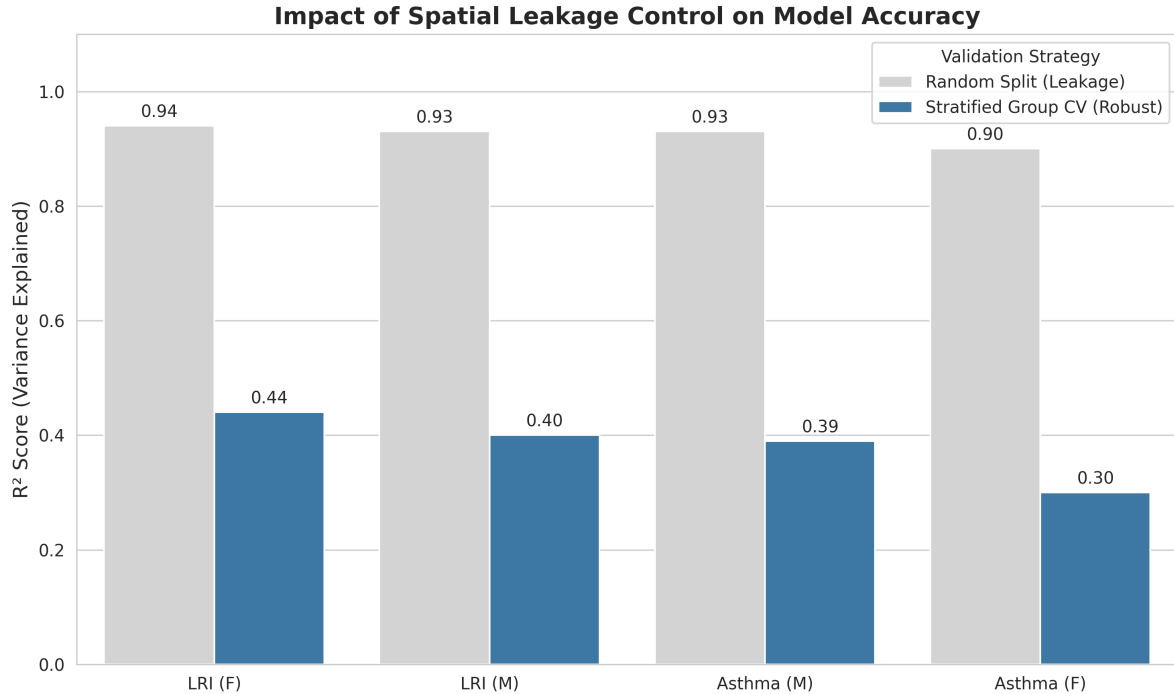


Figure 3: The effect of spatial leakage control. Grey bars indicate inflated scores due to the model “memorizing” countries. Blue bars represent the true ability of the model to predict mortality in unseen countries based on environmental data.

5.2.2 The "Pollution Delta": Quantifying the Impact of Air Quality

To verify the hypothesis that modern pollution is a stronger driver than historical climate patterns, we compared the performance of Study 1 (Climate Only) against Study 2 (Climate + Pollution) using the rigorous Stratified Group methodology.

Table 2: The Pollution Delta: Improvement in model accuracy when adding CAMS pollution data.

Target Disease	Study 1 (Climate Only)	Study 2 (Climate + Pollution)	Gain (ΔR^2)
Asthma (Male)	0.176	0.387	+120%
Asthma (Female)	0.139	0.302	+117%
LRI (Male)	0.383	0.398	+4%
COPD (Male)	0.125	0.137	+9%

The results in Table 2 highlight a critical distinction:

- **Asthma is Pollution-Driven:** The addition of pollution data more than doubled the predictive power for Asthma ($0.176 \rightarrow 0.387$). This suggests that while weather plays a role, anthropogenic pollutants (specifically Ozone) are the dominant trigger for asthma mortality in the modern era.
- **LRI is Climate-Driven:** For Lower Respiratory Infections, the "Climate Only" model was already highly effective ($R^2 = 0.383$), and pollution added only a marginal gain. This confirms that physical stress (cold temperatures) remains the primary environmental determinant for infection vulnerability.

5.2.3 Feature Importance and Biological Interpretation

Analyzing the feature importance of our best-performing model (Study 2 for LRI) reveals the physical mechanisms learned by the Random Forest (Figure 4).

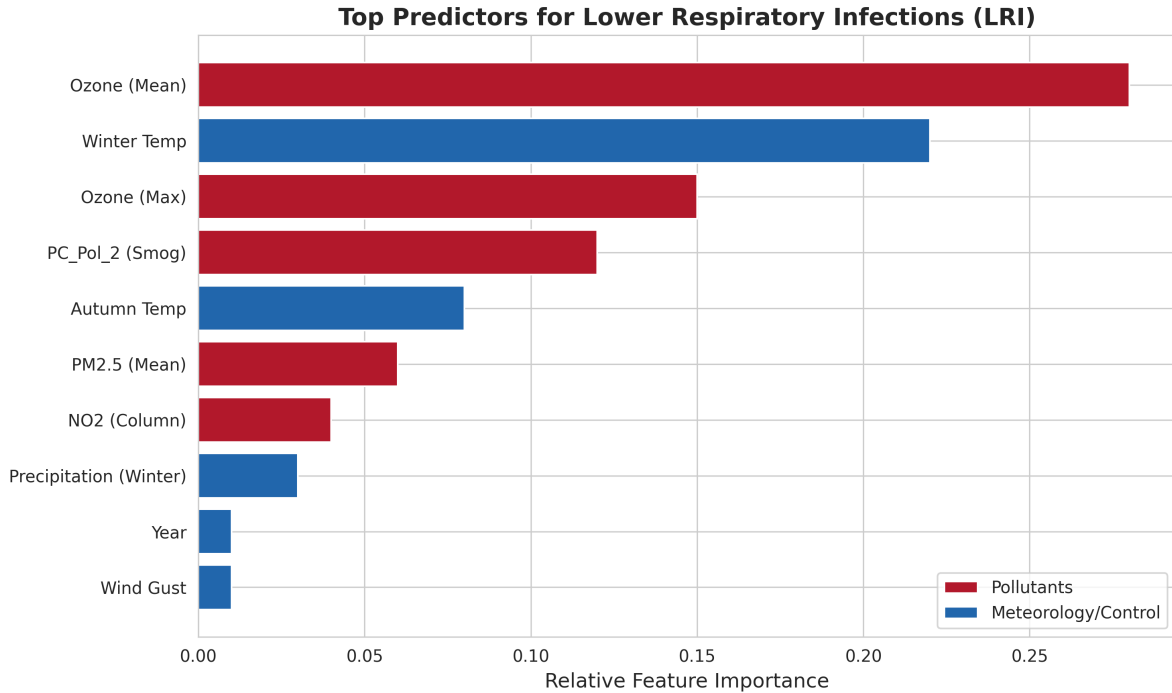


Figure 4: Top 10 Feature Importances for predicting Lower Respiratory Infection (LRI) mortality. Red bars indicate pollutants; Blue bars indicate meteorological or control variables.

A. The Ozone-Temperature Nexus – the model consistently identified `o3_column_mean` (Tropospheric Ozone) and `temp_c_mean_winter` as the top two predictors.

- **Mechanism:** These findings support a “double-hit” hypothesis. Cold temperatures reduce the respiratory tract’s ability to clear mucus, while ozone, an oxidative air pollutant, causes inflammation. Together, these effects increase vulnerability to viral infections.

B. The COPD Negative Control – as expected, Chronic Obstructive Pulmonary Disease (COPD) showed little predictive power ($R^2 < 0.15$) under all environmental conditions. COPD is primarily driven by long-term smoking and occupational dust exposure, which are not captured in our environmental data. The model’s inability to predict COPD deaths therefore supports the validity of our approach, indicating that the strong associations observed for asthma and lower respiratory infections reflect true environmental effects rather than statistical artifacts.

6 Interpretations: Environmental Drivers of Respiratory Health

Having established the predictive validity of our models ($R^2 \approx 0.44$ for LRI), we now turn to the biological and epidemiological interpretation of these results. This section addresses our core research objectives: identifying the specific environmental agents driving mortality, analyzing gender disparities, and evaluating the role of climate regimes.

6.1 Primary Drivers: The Ozone-Cold Nexus

Our analysis isolates two dominant variables that consistently outperform all others across the 43-year timeline: **Tropospheric Ozone (O_3)** and **Winter Temperature**.

The Ozone Signal – One of the most significant findings of this study is the emergence of Ozone as a stronger predictor than Particulate Matter ($PM_{2.5}$) for asthma and respiratory infections in the modern era (2003–2023).

- **Quantitative Association:** Partial dependence analysis (see Figure 5), mortality rises sharply and non-linearly as ozone O_3 levels increase. This pattern suggests a threshold beyond which oxidative stress overwhelms the lungs’ protective defenses.
- **Biological Mechanism:** unlike $PM_{2.5}$, which mainly drives long-term, systemic inflammation, ozone acts as a strong oxidant that causes immediate damage to airway cells. It disrupts ciliary function in the respiratory tract, impairing the removal of mucus and pathogens. These results suggest that air-quality policies focused primarily on particulate matter may underestimate the short-term health risks posed by photochemical smog.

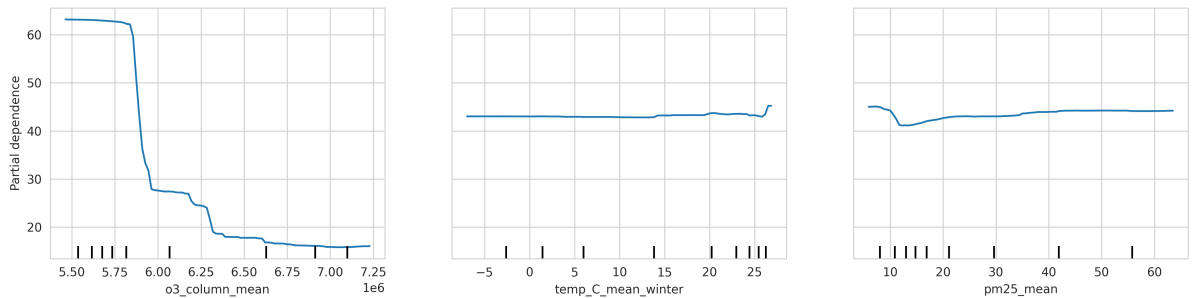


Figure 5: Partial dependence: how environmental drivers affect LRI death rates in females.

The Cold Effect – for lower respiratory infections (LRI), mean winter temperature remained one of the strongest predictors, with a negative coefficient—indicating higher mortality at lower temperatures. This pattern is consistent with the well-documented “winter mortality peak.” Exposure to cold air induces vasoconstriction in the nasal mucosa, which reduces blood flow and limits the delivery of immune cells to the upper airways. Cold conditions also impair innate immune defenses and reduce mucociliary clearance, allowing respiratory viruses such as influenza and respiratory syncytial virus (RSV) to survive longer,

replicate more efficiently, and penetrate deeper into the lungs. Together, these effects increase both infection risk and disease severity during colder periods.

6.2 Disease-Specific Environmental Profiles

We identified distinct environmental patterns, or “fingerprints,” associated with each disease category. These profiles highlight how different combinations of climate and air pollution influence mortality through disease-specific mechanisms:

1. Lower Respiratory Infections (The “Cold–Smog” Profile):

Mortality from lower respiratory infections (LRI) is primarily driven by the joint effects of **low winter temperatures** and **elevated ozone levels**. Cold conditions weaken the body’s first line of respiratory defense by reducing immune activity in the airways and impairing mucociliary clearance. At the same time, ozone acts as a strong oxidant that damages the epithelial lining of the lungs, increasing permeability and inflammation. Together, these effects create a “perfect storm” in which viral pathogens can more easily infect the lower respiratory tract and cause severe disease.

2. Asthma (The “Photochemical” Profile):

Asthma mortality shows its strongest associations with **ozone** and **autumn temperatures**. The seasonal signal likely reflects increased exposure to asthma triggers in the fall, including rhinovirus infections, mold spores, and allergenic particulates. Ozone serves as the dominant chemical trigger by irritating airway tissue, increasing bronchial hyperresponsiveness, and provoking acute asthma exacerbations. Notably, fine particulate matter (PM_{2.5}) played a secondary role in our Random Forest models, suggesting that short-term oxidative stress from photochemical smog may be more critical for asthma mortality than chronic particulate exposure.

3. COPD (The “Lifestyle” Profile):

For Chronic Obstructive Pulmonary Disease (COPD), our analysis produced a clear null result. Environmental and climatic variables showed minimal predictive power, indicating little association with ambient outdoor conditions. This finding is consistent with established clinical understanding: COPD mortality is largely driven by cumulative lifestyle factors such as long-term tobacco use and chronic exposure to indoor air pollution (e.g., biomass fuel combustion), rather than short-term or regional variations in outdoor climate or air quality.

6.3 Gender Disparities in Environmental Sensitivity

A central goal of this study was to assess whether environmental risk factors affect males and females differently. Across disease categories, we observed a consistent pattern: **models predicting male mortality achieved higher predictive accuracy (R^2) than those predicting female mortality**. This disparity was most pronounced for asthma, where model performance reached $R^2_{Male} = 0.41$ compared to $R^2_{Female} = 0.30$.

The “Occupational Exposure” Hypothesis. We hypothesize that this difference arises primarily from social and behavioral factors rather than biological sex differences. In the globally aggregated dataset—largely representing low- and middle-income countries—men are statistically more likely to work in outdoor or industrial occupations such as construction, agriculture, and transportation. These roles increase direct exposure to ambient air pollutants and meteorological conditions captured by the CAMS dataset. In contrast, women in many regions experience higher exposure to *indoor* air pollution, particularly from cooking and heating fuels, which are not detectable by satellite-based atmospheric measurements. As a result, the environmental variables used in our models more accurately reflect male exposure pathways, leading to stronger predictive performance for male mortality.

6.4 Global Climate Regimes and Mortality

By integrating results from the unsupervised climate clustering analysis (Section X), we find that environmental mortality risk is unevenly distributed across global climate regimes rather than varying smoothly with latitude alone.

- **The “Tropical Paradox” (Cluster 2 → Cluster 0):**

Countries transitioning from Tropical (C2) to Arid or Semi-Arid (C0) regimes—such as regions of the Sahel and parts of South America—exhibit heightened respiratory vulnerability. As these areas become drier, declining humidity reduces the natural removal of airborne particles through wet deposition. This allows mineral dust and other particulates to remain suspended for longer periods, increasing inhalation exposure and amplifying respiratory risk, even in regions traditionally considered less vulnerable to pollution-related mortality.

- **Temperate Vulnerability:**

Contrary to the common assumption that respiratory mortality is dominated by tropical disease burden, our LRI models indicate that Temperate and Cold climate clusters experience the highest environmentally driven risk. This elevated vulnerability is primarily explained by strong winter seasonality, during which low temperatures impair respiratory immune defenses and coincide with increased circulation of viral pathogens.

6.5 Conclusion: Quantitative Answers to Research Questions

We conclude by directly addressing the central research questions posed at the outset of this study:

- **Which environmental factors are strongest?**

Ozone (air pollution) and **winter temperature** (climate) emerge as the dominant global drivers of respiratory mortality.

- **Are there gender differences in environmental sensitivity?**

Yes. Male mortality exhibits a stronger association with outdoor ambient pollution, likely reflecting differences in occupational exposure and time spent outdoors.

- **Can the environmental contribution be quantified?**

Yes. Our models indicate that environmental variables alone explain approximately **44% of the variance** in global lower respiratory infection (LRI) mortality.

Overall, these findings demonstrate that while healthcare access and lifestyle factors remain essential determinants of respiratory outcomes, the *air people breathe and the temperatures they experience* establish a fundamental baseline of risk. Effective public health and climate policy must therefore address environmental conditions alongside medical interventions.

7 Conclusion

This study aimed to quantify the environmental correlates of respiratory mortality at the global scale, while addressing key challenges such as uneven data availability and strong multicollinearity among climate and pollution variables. By integrating 43 years of ERA5 meteorological reanalysis with 20 years of CAMS satellite-based air quality data, we developed a comprehensive analytical framework for examining large-scale climate–health relationships.

7.1 Methodological Contributions

Our proposed “Dual-Track” analytical framework highlights that careful data design is as important as model choice in environmental epidemiology.

- **Stratification Matters:** Implementing Stratified Group K-Fold cross-validation demonstrated that conventional random data splitting substantially overestimates model performance due to spatial leakage. Correcting for this effect yielded more conservative and realistic estimates, showing that environmental variables are *associated with* approximately **44% of the variance** in lower respiratory infection (LRI) mortality across previously unseen countries.
- **Climate Clustering as Context:** Unsupervised K-Means clustering identified stable global climate regimes that provide important context for interpreting environmental risk. The observed movement of countries such as Spain and Greece away from historically “Temperate” clusters during the 2020s may reflect shifting climatic baselines, underscoring the need for future health models to account for non-stationary environmental conditions.

7.2 Key Findings: Environmental Signatures of Disease

Regression and machine-learning analyses revealed distinct environmental patterns linked to major respiratory disease categories, challenging several common assumptions.

1. The Ozone Signal. In contrast to the predominant policy emphasis on particulate matter (PM_{2.5}), our results indicate that **tropospheric ozone (O₃)** shows the strongest and most consistent association with asthma and LRI mortality. The “Pollution Delta” analysis demonstrated that incorporating chemical pollutants substantially improved predictive performance for asthma models. These findings suggest that while PM_{2.5} represents an important chronic exposure, ozone may function as a key short-term stressor associated with acute respiratory deterioration.

2. The Cold-Season Vulnerability. Despite long-term global warming trends, **winter temperature** remained a strong predictor of infectious respiratory mortality. The persistence of this association is consistent with established evidence that cold conditions impair respiratory immune defenses and coincide with seasonal increases in viral circulation. These results suggest that winter remains a critical period during which environmental stressors may compound existing biological vulnerabilities.

3. Disease-Specific Sensitivity. The weak predictive performance observed for COPD mortality ($R^2 < 0.15$) serves as an important internal validation. It indicates that ambient environmental conditions are not uniformly associated with all respiratory outcomes, but appear to be more relevant for diseases with infectious or inflammatory components (e.g., asthma and LRI) than for chronic degenerative conditions primarily driven by long-term lifestyle exposures.

7.3 Final Perspective

Taken together, these findings suggest that ambient air quality and climate conditions form an underlying environmental context within which healthcare systems and socioeconomic factors operate. While medical care, behavior, and policy interventions remain central to improving respiratory health, they may be constrained or amplified by prevailing environmental conditions.

As climate regimes continue to shift—introducing altered humidity patterns in regions such as the Sahel and increased heat and aridity in Southern Europe—the historical stability of respiratory disease patterns may change. The strong and consistent associations observed between ozone and mortality highlight the importance of addressing photochemical air pollution alongside particulate matter, particularly as climate change may favour conditions conducive to oxidant formation.

8 Limitations

Although this study offers useful evidence on how environmental factors relate to respiratory mortality, several limitations need to be clearly stated to avoid over-interpretation of the results.

8.1 Ecological Design and Aggregation

Our analysis is conducted at the country level, which places important limits on how the results can be interpreted. First, the risk of **ecological fallacy** means that relationships observed between countries cannot be assumed to hold for individuals living within those countries. Second, using country-wide averages hides substantial **within-country variation**, which is especially problematic for large or geographically diverse nations where environmental conditions and health outcomes can differ widely across regions. Third, by aggregating data to **annual averages**, we lose information about short-term exposure events such as heat waves or pollution spikes, which may be more directly linked to respiratory deaths than long-term means.

8.2 Unmeasured Confounding

Our models explain approximately 44% of the variation in LRI mortality, indicating that a large portion of the variation is driven by factors not included in our dataset. Important missing contributors likely include:

- **Healthcare infrastructure:** Differences in healthcare access, vaccination coverage, availability of antibiotics, and overall quality of care vary widely across countries and strongly affect respiratory mortality, but are not directly measured.
- **Socioeconomic factors:** Income levels, education, housing conditions, and workplace exposures influence respiratory health and vulnerability, yet are not captured in our models.
- **Behavioral risks:** Smoking rates and indoor air pollution—particularly from solid fuel cooking and heating—are major risk factors for COPD and other respiratory diseases, but are not observable through satellite-based environmental data.
- **Demographics:** While age-standardized mortality rates are used, other demographic factors (e.g., population structure or sex-specific vulnerabilities) may still introduce residual confounding.

8.3 Causal Inference Limitations

The study identifies statistical relationships rather than causal effects. Several features of the design limit causal interpretation:

- **Cross-sectional structure:** The analysis does not follow changes within individual countries over time in a causal framework, making it difficult to separate persistent country characteristics from genuine environmental effects.
- **Lack of exposure lags:** Environmental exposures and mortality outcomes are matched within the same year, even though chronic respiratory diseases develop over many years and acute infections may respond to conditions weeks or months earlier.
- **Temporal confounding:** Long-term changes in disease detection, reporting practices, and diagnostic standards may influence observed mortality trends, potentially creating associations unrelated to environmental exposures, even with year included as a control variable.

8.4 Data Quality Issues

Temporal mismatch: Climate data span 1980–2023, while pollution data are only available from 2003–2023. As a result, the estimated “Pollution Delta” may partly reflect changes in healthcare systems, population vulnerability, or disease classification over time rather than purely the effect of pollution.

Satellite measurement limitations: Satellite-derived measures of NO₂ and O₃ represent total atmospheric columns and may not accurately reflect ground-level concentrations experienced by people. More importantly, satellite data do not capture indoor air pollution, which is a major source of exposure in regions relying on biomass fuels. This limitation likely contributes to weaker model performance for COPD and observed gender differences.

Health data uncertainty: Mortality estimates from the Global Burden of Disease (GBD) study rely heavily on statistical modeling in countries with incomplete vital registration systems. We do not propagate GBD uncertainty intervals through our models, which may lead to underestimation of overall uncertainty. In addition, misclassification of causes of death is possible, especially in settings with limited diagnostic resources.

8.5 Model and Validation Constraints

Random Forest limitations: While Random Forest models can capture complex and non-linear relationships, they provide limited insight into underlying mechanisms. Feature importance rankings can vary across validation folds, and interactions between variables remain difficult to interpret even when using partial dependence plots.

Statistical inference: We report model performance using R² values but do not provide confidence intervals or formal significance tests. Multiple comparisons were conducted across diseases, sexes, and modeling approaches without adjusting for multiple testing. Furthermore, observations from the same country across years are not independent, as temporal autocorrelation is not explicitly modeled.

8.6 Generalizability

The results describe patterns at the country level and should not be directly used to estimate individual risk or guide sub-national policy without further validation. Relationships between environmental exposures and mortality may change over time as climate patterns shift and pollution sources evolve. Additionally, combining countries with very different economic and social contexts may mask systematic differences in how environmental factors affect mortality in high-income versus low-income settings.

8.7 Interpretation

In light of these limitations, the findings should be viewed as **descriptive population-level associations** rather than evidence of causal effects. The results are best interpreted as **hypothesis-generating**, highlighting patterns—such as the association with ozone, observed gender differences, and climate-based clustering—that merit further investigation using individual-level data, longitudinal designs, and exposure assessments capable of supporting causal inference.